Arcade : Developers documentation

Here you will find all the information about how the arcade works in relation to the display and games libraries.

1 - How does Arcade works

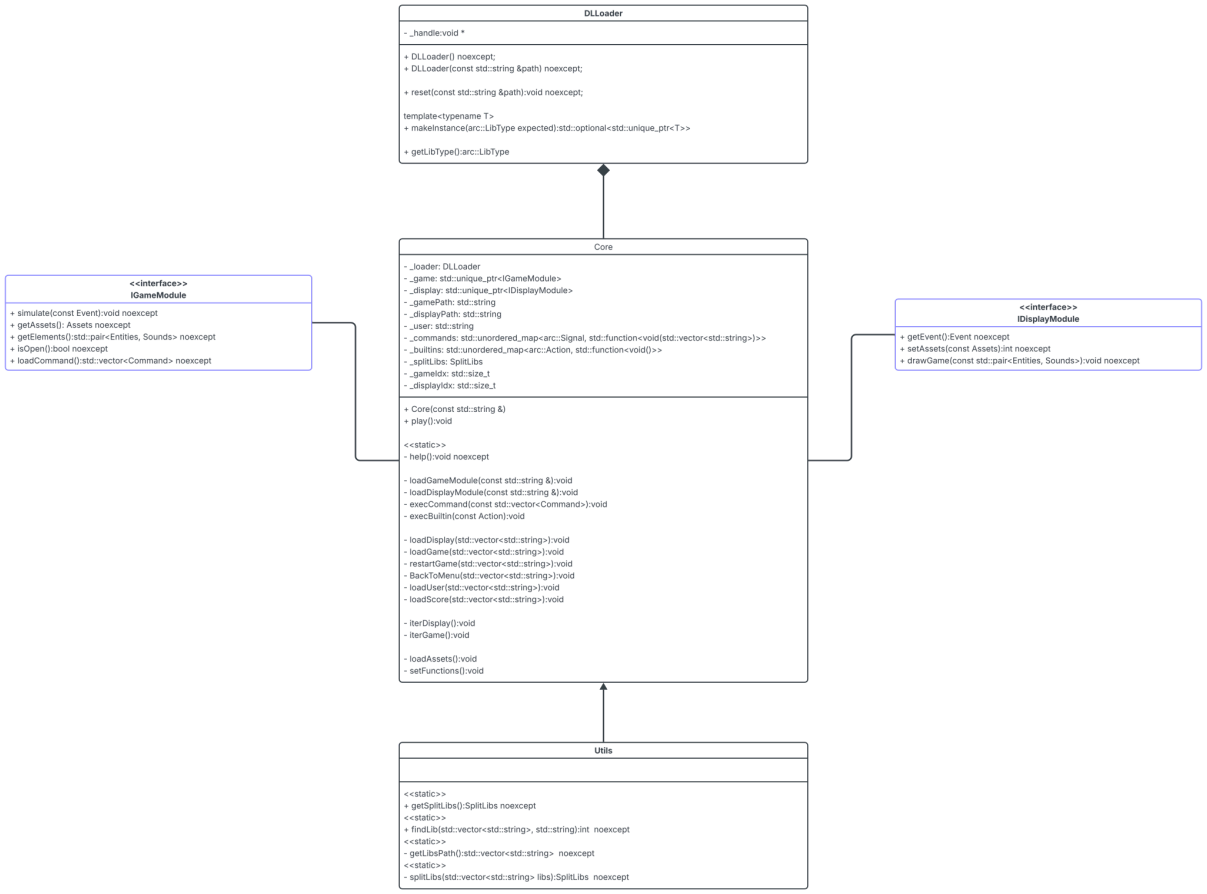
The goal of Arcade is to allow the same game to run using different graphics dynamic libraries. To achieve this, the program is divided into 3 parts:

- The **graphics** part, which handles rendering sprites at the correct positions.
- The **game** part, which handles calculating interactions between entities and their positions (relative to the game).
- The **core** part, which links the graphics and game sections and notably contains the main game loop.

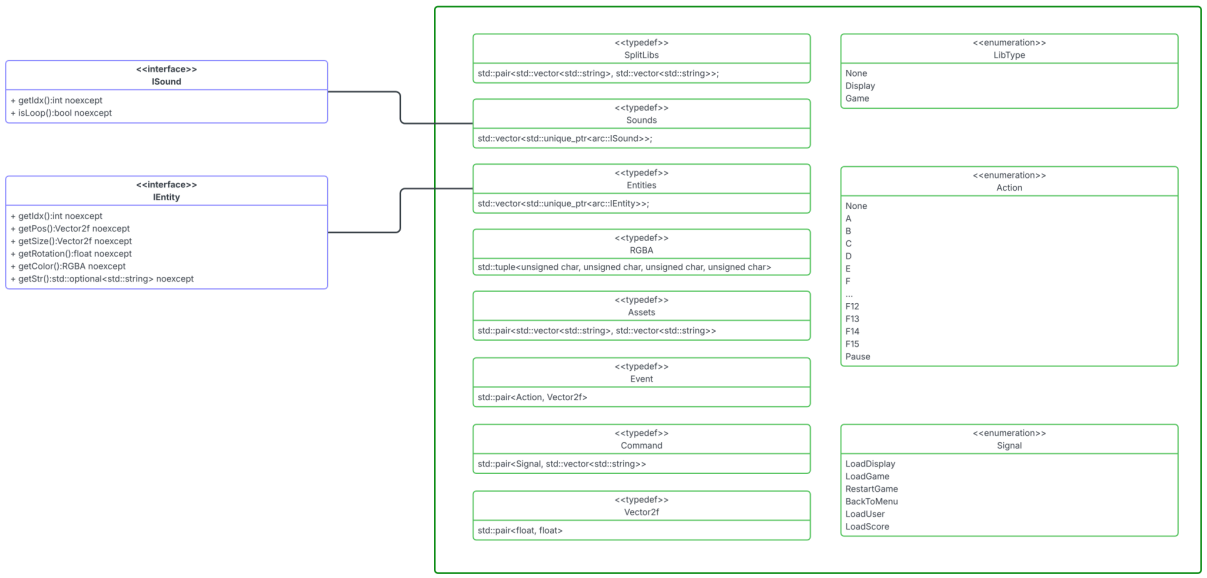
You do not need to modify the "core" part to implement new games and/or new graphics libraries.

To make the project architecture clearer, we have provided an UML diagram below containing all our interfaces and classes, helping you better understand how they interact with one another.

Here is the UML diagram of the **Arcade** architecture:



Which uses the following dependencies:



2 - How does the core communicate with the libraries

Firstly, at the start of the program and using the DLLoader (encapsulation of **dl** library), we create two lists of loaders (games and graphics libraries), which allows us to load as we want any game of display instances at any moment of the program execution.

After this part of **initialisation**, we can use the program parameter to open a graphical instance with a **menu game library**.

Then, using the *getAssets* and *setAssets* we communicate the required assets to the graphical library, which then can use these to load the assets with its required format. This allows the graphics library to choose format while the game chooses the files names.

Right after this handshake, the core start to enter a loop to, firstly, get the entities from the game and them to the graphical library to draw them, and then, handle events, which can be keyboard and mouse, or commands that the game makes to the core (for example, changing library or restarting the game).

3 - Contributing to the Arcade

If you would like to contribute to the Arcade project, you can look after **GameDocumentation.pdf** and **GraphicDocumentation.pdf** in the doc folder.